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Improving structural heart disease screening: AI-ECG and novice AI-guided focused cardiac ultrasound



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Structural heart disease (SHD) causes substantial morbidity, yet early detection is limited. In this prospective study, AI-enhanced ECG and AI-guided focused cardiac ultrasound (AI-FoCUS) were assessed for detecting SHD (left ventricular EF < 40%, ≥moderate aortic stenosis, hypertrophic cardiomyopathy, or cardiac amyloidosis) in 995 adults against comprehensive echocardiography performed within 30 min of AI-FoCUS. Nine complete novices used handheld ultrasound (Philips Lumify) with real-time AI guidance (UltraSight) with images interpreted by blinded experts. AI-FoCUS was feasible in 97.3%. Prevalence was 5.0% for EF < 40%, 3.5% for ≥moderate aortic stenosis, and 8.8% for hypertrophic cardiomyopathy or amyloidosis. AI-ECG alone yielded NPV 95.8% but PPV 29.6%, limiting standalone screening. AI-FoCUS alone achieved PPV 95.4% and NPV 98.0%, however requires imaging in all. A two-step strategy (positive AI-ECG followed by AI-FoCUS) maintained high PPV/NPV (96.1%/95.6%) but limited FoCUS imaging to 47%.

In this echo referral cohort, novice AI-guided FoCUS acquisition with blinded expert interpretation was feasible and accurate for detecting selected SHD phenotypes. A two-step AI-ECG then AI-FoCUS strategy reduced FoCUS utilization, but lowered sensitivity compared with AI-FoCUS for all. However, both FoCUS-based strategies were superior to AI-ECG based screening alone. This work requires further assessment and validation in lower prevalence populations.

Cardiovascular (CV) disease, including structural heart disease (SHD), is highly prevalent and remains a leading cause of death in the United States and the European Union¹. In 2020 the economic burden of cardiovascular disease in the US was \$400 billion, and this is projected to triple to \$1490 billion by 2050, underscoring the urgent need for the implementation of low-resource screening programs and preventative medicine¹. The most common CV diseases such as reduced left ventricular ejection fraction (EF) and aortic valve stenosis (AS), are often preceded by a phase of asymptomatic structural impairment^{2–10}. Early detection and diagnosis of SHD are necessary for effective intervention and improved survival^{2–12}. Yet, patients with asymptomatic SHD often go undetected and hence miss the benefits of early intervention.

Currently, there is no cost-effective screening method for SHD. Formal echocardiography performed by trained sonographers is too resource- and time-intensive for screening asymptomatic populations. Emerging

literature supports artificial intelligence-enhanced 12-lead electrocardiography (AI-ECG) as a non-invasive, scalable, and inexpensive alternative test. A convolutional neural network (CNN) applied to an ECG detects subtle patterns in electrical signals, enabling detection of SHD conditions such as moderate or greater AS, left ventricular systolic dysfunction (LVSD), hypertrophic cardiomyopathy, and cardiac amyloidosis^{13–16}. Despite considerable optimism regarding this technology due to very high negative predictive values, lower specificities and positive predictive values may limit the use of AI-ECG as a standalone tool due to false positives and unnecessary referrals for diagnostic testing such as echocardiography.

Focused cardiac ultrasound (FoCUS) with AI-guidance software is emerging as a method for less-experienced users (or potentially users without any clinical or imaging experience) to capture basic cardiac images¹⁷. The AI guides the user in real time to position the ultrasound

probe, optimize the image, and indicates when an optimal image is present to allow for the ability to obtain high quality cardiac ultrasound images. Compared to a formal echocardiogram performed by trained sonographers, this technology is relatively inexpensive, accessible, and potentially scalable for SHD screening in resource-limited settings. However, due to limited clinical data, it is unclear whether novice users guided by AI can obtain satisfactory FoCUS images for SHD screening. It is also uncertain whether these images can be effectively integrated with AI-ECG to reduce false positives and improve diagnostic accuracy.

Our study hypothesized that the addition of AI-guidance would allow individuals without any prior imaging or clinical experience (truly novice users) to obtain AI-FoCUS to accurately detect abnormalities of SHD. The aim was to evaluate the diagnostic performance of the AI-FoCUS scan to identify either (i) LVSD, (ii) moderate or greater AS, (iii) hypertrophic cardiomyopathy or (iv) cardiac amyloidosis. We secondarily assessed the diagnostic performance of the Mayo Clinic AI-ECG algorithms to identify the same 4 pathologies as well as a two-step combined test where the 4 AI-ECG algorithms were first applied, and if positive for one pathology, then the results of the AI-FoCUS scan were considered.

Results

Study cohort

The cohort comprised 995 prospectively enrolled patients who underwent an AI-FoCUS within 30 days of an AI-ECG. They also all underwent a comprehensive diagnostic echocardiogram within 30 min of their FoCUS (Fig. 1). The cohort included individuals with a broad spectrum of left ventricular pathology, as well as various forms of valvular heart disease (Table 1). All patients underwent a 12-lead AI-ECG within 30 days of the AI-FoCUS and echocardiogram, and the majority (84%) had the AI-ECG within 48 h (median of 0, IQR of 0–1). As detailed in Table 1, the cohort exhibited considerable variability in age, weight, body mass index (BMI), and echocardiographic findings. The median age was 69 years (interquartile range: 58–77), with females accounting for 39% of the population. The mean BMI was $28 \pm 6 \text{ kg/m}^2$, with values ranging from 17 kg/m^2 to 58 kg/m^2 .

Feasibility of AI-FoCUS scan

Among the 995 patients in the study, 27 (2.7%) individuals had AI-FoCUS examinations in which no diagnostic images could be obtained. In the remaining cohort of subjects with feasible AI-FoCUS studies, the median scan duration was 4 min (interquartile range: 3–7 min). Image quality and feasibility varied across the six standard imaging views, as illustrated in Fig. 2, data summarized in Table 2. The parasternal long-axis (PLAX) and

parasternal short-axis (PSAX) views demonstrated the highest feasibility, followed by the apical four-chamber (A4C) view, which was moderately feasible. The apical three-chamber (A3C) and apical two-chamber (A2C) views were the least feasible. Interobserver agreement on quality, in 290 random studies reviewed by both experts, was high ($\kappa = 0.93$).

The operators' subjective assessment of scan difficulty was significantly correlated with both image feasibility and quality (Fig. 3). Scans perceived as “easy” (25.3%) yielded good or excellent quality images in 69–88% of parasternal and 65–82% of apical views. In contrast, scans rated as “moderate” difficulty (42.1%) and “hard” (33.3%) resulted in good or excellent image quality in only 48–72% (moderate) and 32–64% (hard) of parasternal views, and 47–65% (moderate) and 17–34% (hard) of apical views (all $p < 0.0001$).

LV EF classification by AI-ECG and AI-FoCUS

Both the AI-ECG and the AI-FoCUS were targeted to assess for a threshold of an LVEF $< 40\%$. Of the 995 patients, 50 had a LVEF $< 40\%$ for a prevalence of 5.0% by standard comprehensive echocardiography. The AI-ECG had a sensitivity of 86.0% and a specificity of 88.6% for the detection of an LVEF $< 40\%$. Of the 108 false positive AI-ECGs, 41 had an LVEF between 40% and 49%, and 67 an LVEF $\geq 50\%$ on comprehensive echocardiography. In contrast, only 2 AI-FoCUS scans were false positive, and both had LVEF 40–49% and were near the diagnostic threshold. The AI-FoCUS had a sensitivity of 92.0% and a specificity of 99.9% for the detection of an LVEF $< 40\%$. For a LVEF $< 40\%$, the PPVs and NPVs were 28.5% and 99.2% for the AI-ECG and 97.9% and 99.6% for the AI-FoCUS (Table 3).

Aortic stenosis classification by AI-ECG and AI-FoCUS

Both the AI-ECG and the AI-FoCUS were targeted to assess for a threshold of moderate or greater AS. 108 patients of the 995 had an aortic valve prosthesis. These patients were excluded from the AS analysis. Of those patients with a native aortic valve, 35 had moderate or greater AS (14 moderate, 7 moderate-severe, and 14 severe) for a prevalence of 3.9%. The AI-ECG had a sensitivity of 68.6% and a specificity of 69.5% for the detection of moderate or greater AS. Of the 262 false positive AI-ECG analyses for AS, 20 had mild AS and 242 did not have AS. AI-FoCUS reduced these to just 3 false positives, all of whom had mild AS by reference echocardiography. The AI-FoCUS had a sensitivity of 94.3% and a specificity of 99.6% for the detection of moderate or greater AS. The PPVs and NPVs were 8.4% and 98.2% for the AI-ECG and 91.7% and 99.7% for the AI-FoCUS (Table 3).

Fig. 1 | Study flow of the study participants. 995 patients were recruited within 30 days of having an AI-ECG. All underwent a focused cardiovascular ultrasound performed by one of 9 novice users guided by AI software (AI-FoCUS) and a diagnostic echocardiogram. Of the 995 AI-FoCUS scans, 97.3% were technically feasible; 2.7% ($n = 27$) were not and hence excluded. Created in BioRender. Kane, G. (2026).

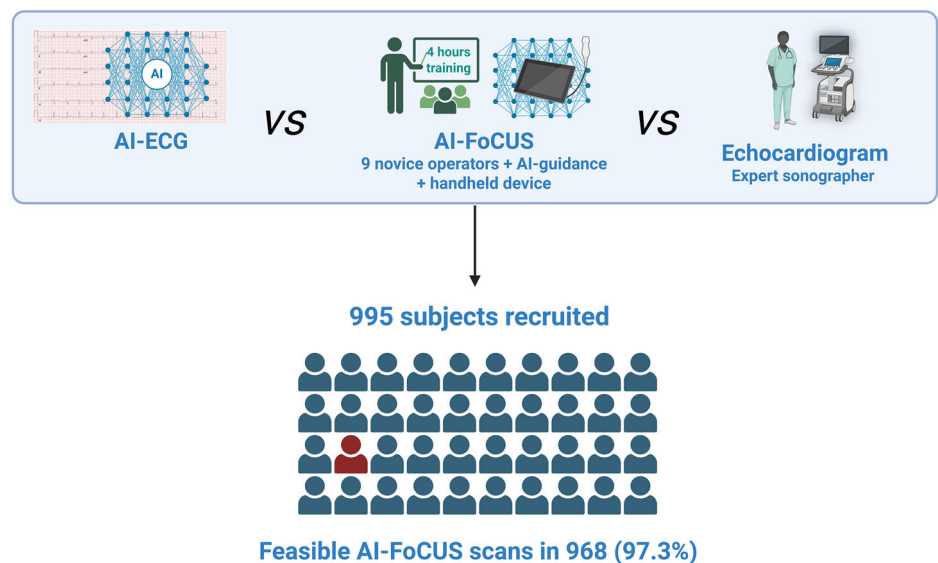


Table 1 | Study population (*n* = 995)

Clinical demographics	
Age, years	69 (IQR 58–77); range 18 to 93
Sex, male	609 (61%)
Race and ethnicity	
• White	952 (95.7%)
Not Hispanic or Latino	912
Hispanic or Latino	20
Unknown	20
• Black or African American	23 (2.3%)
• Asian	12 (1.2%)
• American Indian, Alaska Native, Native Hawaiian or Other Pacific Islander	5 (0.5%)
• Unknown / not reported	2 (0.2%)
Height, cm	173 ± 10; range 140 to 203
Weight, kg	85 ± 19; range 41 to 166
Body mass index, kg/m ²	28.5 ± 5.7; range 17 to 58
Body mass index > 30 kg/m ²	323 (32.5%)
Systolic blood pressure, mm Hg	131 ± 19; range 89 to 196
Diastolic blood pressure, mm Hg	77 ± 10; range 42 to 113
Heart rate, bpm	68 ± 13; range 35 to 131
Rhythm	
• Sinus	827, 87%
• Atrial fibrillation/flutter	112, 11%
• Paced	22, 2%
Indication for echocardiogram	
• Dyspnea, known or suspected cardiomyopathy, or heart failure	253, 25.4%
• Palpitations, known or suspected arrhythmia	223, 22.4%
• Murmur, known or suspected valvular disease	196, 19.7%
• Pre or post chemotherapy	64, 6.4%
• Known or suspected coronary artery disease	58, 5.83%
• Lung disease, known or suspected pulmonary hypertension	53, 5.3%
• Vascular disease or aortic dilation	49, 4.9%
• Other, including prior heart transplant and pericardial disease	99, 10.0%
Echocardiographic findings	
LV size, enlarged	219, 22.0%
LV size, moderate or greater enlarged	93, 9.3%
LV volume indexed to BSA, ml/m ²	71 ± 19
LV ejection fraction, %	58 ± 9
LV EF, less than 50%	127, 12.8%
LV EF, less than 40%	50, 5.0%
LV mass indexed to BSA, mg/m ²	94 ± 31
Right ventricular dysfunction, > mild	66, 6.6%
Right ventricular systolic pressure, mm Hg	33 ± 11
Pericardial effusion more than tiny	23, 2.3%
Aortic valve prosthesis	101, 10.2%
≥ moderate aortic valve stenosis	35, 3.5%

Table 1 (continued) | Study population (*n* = 995)

Clinical demographics	
Hypertrophic cardiomyopathy	63, 6.3%
Cardiac amyloidosis	25, 2.5%
Presence of any of the following:	
(i) EF < 40%; (ii) ≥ moderate aortic stenosis; (iii) hypertrophic cardiomyopathy or (iv) cardiac amyloidosis	163, 16.4%

Values are presented as mean ± standard deviation (if normally distributed) or median and interquartile range (IQR) and percents where appropriate. The total range is also given.

EF ejection fraction, LV left ventricle, RV right ventricle.

HCM/amyloidosis classification by AI-ECG and AI-FoCUS

Of the 995 patients, 87 had a formal diagnosis of either hypertrophic cardiomyopathy (*n* = 63) or cardiac amyloidosis (*n* = 25) for a prevalence of 8.8%. The AI-ECG had a sensitivity of 82.8% and a specificity of 77.3% for HCM or amyloid. The AI-FoCUS had a sensitivity of 84.9% and a specificity of 99.4% for HCM or amyloid. For either HCM or cardiac amyloid, the PPVs and NPVs were 25.9% and 97.9% for the AI-ECG and 93.6% and 98.6% for the AI-FoCUS (Table 3). ROC curves with 95% confidence intervals for each of the AI-ECG outputs are shown in Supplementary Fig. 1 and Calibration plots in Supplementary Fig. 2.

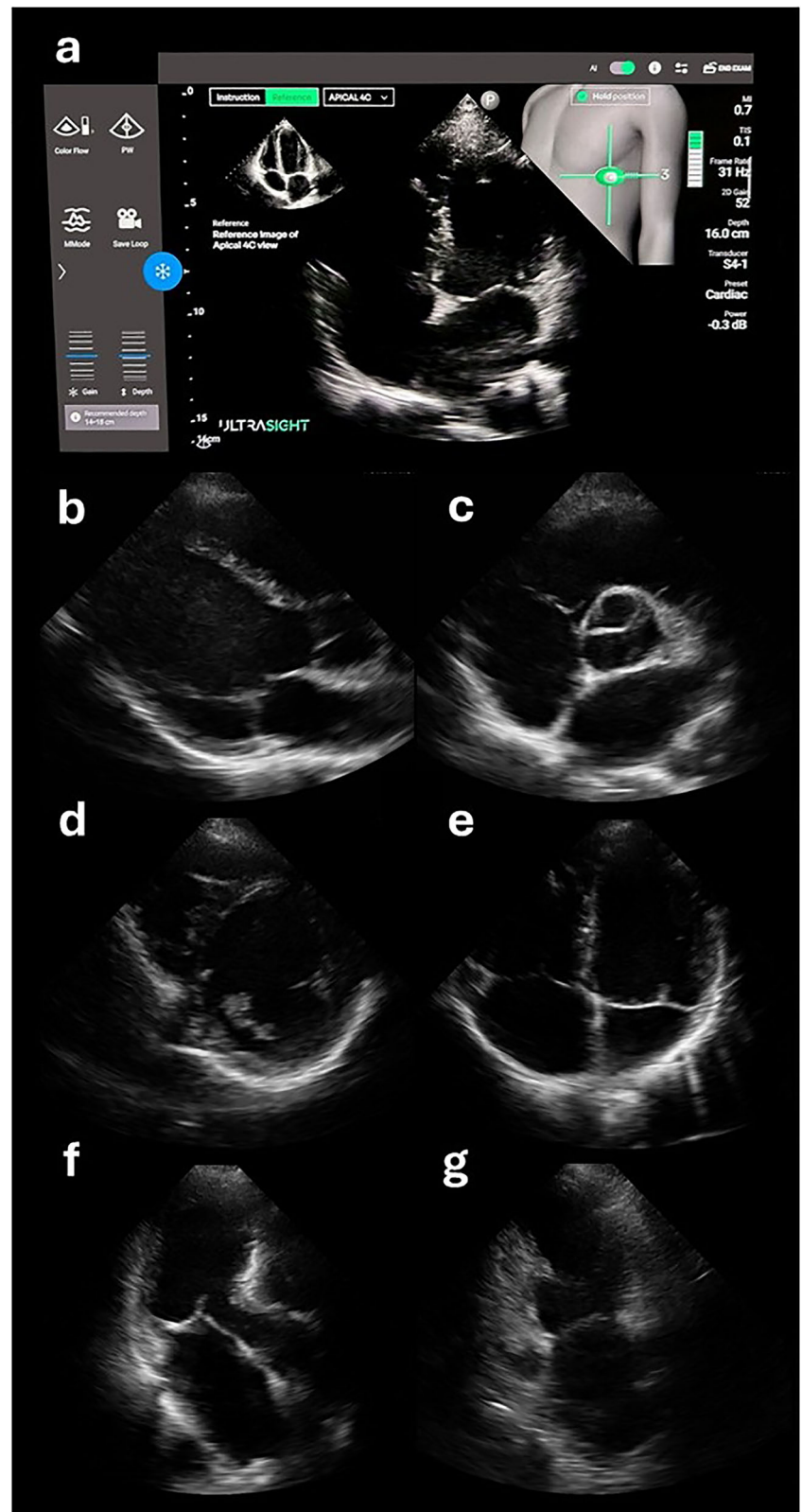
Composite structural heart disease classification by AI-ECG and AI-FoCUS

Patients were then evaluated for the presence of any of the 4 variables (low EF, moderate or greater AS, hypertrophic cardiomyopathy, or cardiac amyloidosis). Comparison of the results is demonstrated in a side-by-side confusion matrix plot (Fig. 4). Of the 995 patients, 163 had one of these four diagnoses for an overall prevalence of 16.4%. The results are outlined in Table 4.

The prevalence of SHD was not different between those patients scanned within 48 h (<2 days) of their ECG or 2–30 days between ECG and FoCUS. Furthermore, in a sensitivity analysis restricting AI-ECG to those obtained within 48 h, performance estimates were essentially unchanged, supporting minimal bias from the longer screening window. Operator-level image quality varied across novice imagers for both the apical 4-chamber and parasternal long-axis views. For the apical four chamber view, the distribution of poor/missing, adequate, and good/excellent images differed by operator ($\chi^2 = 40.17$, *df* = 16, *p* = 0.0007). For the parasternal long axis view, between-operator heterogeneity was more pronounced ($\chi^2 = 68.42$, *df* = 16, *p* < 0.0001). However, study subject BMI also differed across operators, a factor known to be associated with image quality. After adjustment for BMI, difference in operator remained associated with image quality (likelihood-ratio $\chi^2 = 71.7$, *df* = 16, *p* < 0.0001), and higher BMI was independently associated with lower image quality (likelihood-ratio $\chi^2 = 33.8$, *df* = 2, *p* < 0.0001). Non-diagnostic AI-FoCUS scan rates ranged from 0.0% to 4.8% across the nine novice imagers, with an overall non-diagnostic rate of 2.7% (27/995). There was no significant difference between operators in non-diagnostic scan rate (Fisher–Freeman–Halton exact test, *p* = 0.70).

We also assessed the model diagnostic accuracy in the subgroup of patients in whom SHD was not previously established (*n* = 783; prevalence 6.1%). AI-ECG alone yielded sensitivity 87.5%, specificity 60.5%, PPV 12.7%, and NPV 98.7%; AI-FoCUS yielded a sensitivity of 89.4%, specificity 98.9%, PPV 84.0%, and NPV 99.3%. In prevalence-adjusted scenario analyses of all subjects, the PPV of AI-ECG alone declined substantially as assumed prevalence decreased, whereas the FoCUS-based strategies retained markedly higher PPV under the assumption of stable operating characteristics (Fig. 5). At an assumed SHD prevalence of 5%, the estimated PPV was 10.4% for AI-ECG (NPV of 98.9%) and 84.5% for AI-FoCUS (NPV of 99.5%) and at 2% prevalence, the corresponding PPV estimates were 4.3% and 67.9% respectively for AI-ECG and AI-FoCUS with NPV 99.6% and 99.8% (Fig. 5).

Fig. 2 | AI-FoCUS example. A screenshot of the UltraSight software, which depicts the AI-visual guide to users for probe position and image quality (a). The standard AI-FoCUS exam included the following six views: three from the parasternal acoustic window: **b** parasternal long axis view, **c** parasternal short axis at the aortic valve level, and **d** parasternal short axis at the mid-ventricular level, as well as three from the apical window with standard views of: **e** the apical 4 chamber, **f** 3 chamber, and **g** 2 chamber views.



We then tested the diagnostic accuracy of a simulated two-step diagnostic screening process where the results of the AI-FoCUS were only included in those patients who had an abnormal AI-ECG. i.e., if the AI-ECG was negative for all four SHD variables, then the study was deemed negative. If the AI-ECG was positive for any of the four algorithms, then the results of the AI-FoCUS were applied (Fig. 6). This cohort excluded all subjects

without interpretable AI-FoCUS images to allow for direct comparison between screening models. For the composite SHD endpoint in participants with interpretable AI-FoCUS scans ($n = 968$; SHD prevalence 16.6%), AI-ECG alone showed a reasonably high rule-out performance but limited rule-in performance (sensitivity 87.0%, specificity 60.7%, PPV 30.6%, NPV 95.9%). Performing AI-FoCUS for all subjects substantially improved

Table 2 | Quality of images obtained with AI-FoCUS

	Views					
	PS LA	PS AV	PS MV	A4C	A3C	A2C
Excellent	97, 9.7%	23, 2.3%	113, 11.4%	29, 2.9%	13, 1.3%	13, 1.3%
Good	570, 57.3%	456, 45.8%	533, 53.6%	559, 56.2%	424, 42.6%	383, 38.5%
Adequate	231, 23.2%	339, 34.1%	221, 22.1%	244, 24.5%	318, 32.0%	315, 31.7%
Poor	64, 6.4%	107, 10.8%	63, 6.3%	53, 5.3%	67, 6.7%	78, 7.8%
Missing	33, 3.3%	70, 7.0%	65, 6.5%	110, 11.0%	173, 17.4%	206, 20.7%

AI-FoCUS artificial intelligence-guided focused cardiovascular ultrasound, PS LA parasternal long axis, PS AV parasternal short axis at the aortic valve level, PS MV parasternal short axis at the mid left ventricular level, A4C apical 4 chamber view, A3C apical 3 chamber view, A2C apical 2 chamber view.

Fig. 3 | Relationship of user perceived difficulty with expert-rated image quality. Demonstrated in scatterplot format are the proportion of scans grouped by the perceived difficulty by the user (x-axis) against the image quality as graded by the expert (y-axis) for the parasternal long axis (a); parasternal short axis of the aortic valve level (b); and at the mid-ventricular level (c); and the apical 4-chamber (d); 3-chamber (e) and 2-chamber (f).

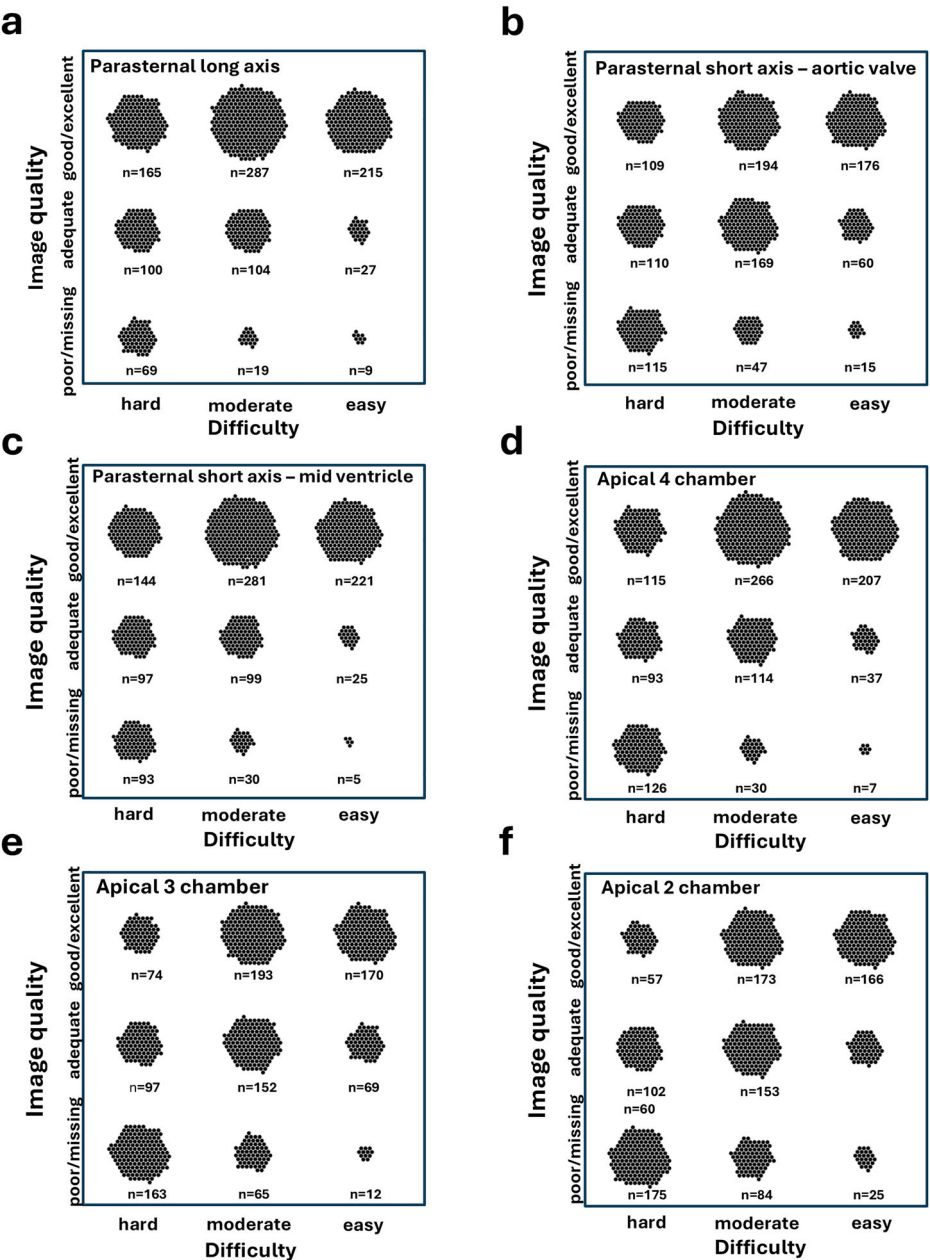
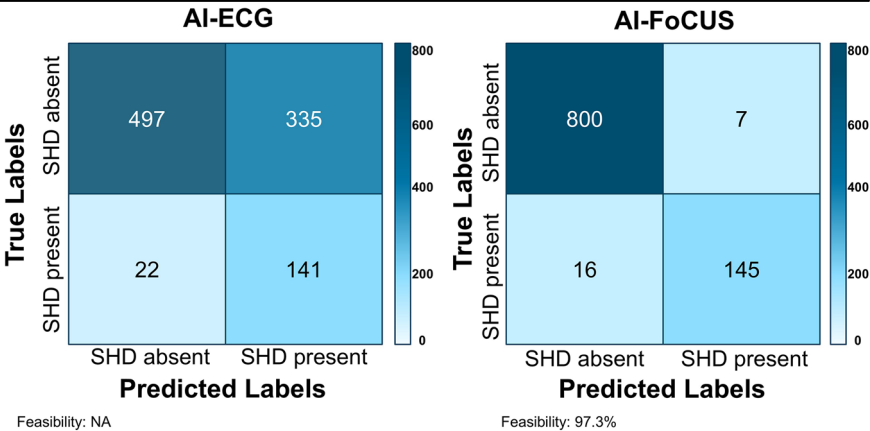


Table 3 | Measures of diagnostic accuracy of the AI-ECG and AI-FoCUS for the detection of SHD

	Sensitivity	Specificity	PPV	NPV	+ve LR	−ve LR	Accuracy
AI-ECG							
• Low EF	0.860	0.886	0.285	0.992	7.525	0.158	0.884
• AS	0.686	0.695	0.084	0.982	2.246	0.452	0.694
• HCM/amyloid	0.828	0.773	0.259	0.979	3.648	0.223	0.778
AI-FoCUS							
• Low EF	0.920	0.999	0.979	0.996	854.68	0.080	0.995
• AS	0.943	0.996	0.917	0.997	251.11	0.057	0.994
• HCM/amyloid	0.849	0.994	0.936	0.986	151.09	0.152	0.982

AI artificial intelligence, AS aortic stenosis, ECG electrocardiogram, EF ejection fraction, AI-FoCUS AI-guided focused cardiac ultrasound, HCM hypertrophic cardiomyopathy, PPV positive predictive value, NPV negative predictive value, +ve LR positive likelihood ratio, −ve LR negative likelihood ratio, SHD structural heart disease.

Fig. 4 | Side-by-side confusion matrix plots. Demonstrated are the results for both the AI-ECG on the left and the AI-FoCUS on the right for the prediction of the composite structural heart disease endpoint against the ground truth of the echocardiogram. Note by study design all patients had an AI-ECG. The feasibility of the AI-focus was 97.3%; 2.7% ($n = 27$) were not feasible and hence excluded.



diagnostic classification (sensitivity 90.1%, specificity 99.1%, PPV 95.4%, NPV 98.0%). In the simulated two-step strategy, AI-FoCUS utilization decreased to 457/968 (47.2%) subjects while maintaining high specificity and PPV at the expense of reduced sensitivity (sensitivity 77.0%, specificity 99.4%, PPV 96.1%, NPV 95.6). Relative to AI-FoCUS for all, the two-step strategy reduced FoCUS utilization by 511 examinations (52.8%) but reduced true-positive detection by 21 cases while decreasing false-positive final classifications by only two. This corresponds to an incremental 24 additional AI-FoCUS scans per extra SHD case detected if AI-FoCUS were performed universally (Table 4). The 21 incremental false negatives were all attributable to the initial AI-ECG gating step and comprised of one case of LVEF < 40%, ten cases of $\geq$ moderate AS, eight cases of HCM, and two cases of cardiac amyloidosis, without overlap. Among these incremental false negatives, no patients had LVEF < 35%, six had severe AS, and fifteen represented previously established SHD (Supplementary Table). The remaining 16 false negatives were shared by both AI-FoCUS for all and the two-step strategy and therefore reflect limitations of limited-view FoCUS rather than the AI-ECG.

Interobserver agreement was assessed in a subset of patients ($n = 290$) where a 2nd blinded expert reviewed the FoCUS images and scored for the presence of SHD. The agreement between observers was excellent with a Kappa of 0.94 (95% CI 0.87–0.99).

As a preliminary decision-analytic assessment, we performed standard decision curve analysis for the composite SHD endpoint in the common interpretable cohort ($n = 968$); Fig. 7. Across threshold probabilities of 2% to 20% for referral to comprehensive echocardiography, AI-FoCUS for all provided the highest net benefit among the three tested screening strategies. At threshold probabilities of 5%, 10%, and 20%, net benefit was 0.147 (95% CI 0.126–0.170), 0.147 (95% CI 0.125–0.169), and

0.145 (95% CI 0.124–0.168) for AI-FoCUS for all; 0.128 (95% CI 0.108–0.148), 0.128 (95% CI 0.108–0.148), and 0.127 (95% CI 0.107–0.147) for the two-step strategy; and 0.127 (95% CI 0.106–0.150), 0.108 (95% CI 0.086–0.131), and 0.063 (95% CI 0.038–0.088) for AI-ECG alone. The two-step strategy and AI-ECG alone were approximately equivalent at a 5% threshold, whereas the two-step strategy showed higher net benefit at thresholds of 10% and 20%. These findings suggest that the principal advantage of the two-step strategy is reduced AI-FoCUS utilization rather than superior decision-analytic performance relative to universal AI-FoCUS.

Discussion

As the population ages with an increasing prevalence of cardiovascular risk factors such as hypertension and diabetes, the global burden of SHD rises¹. Early detection of SHD may provide an opportunity for interventions, including risk factor modification or pharmacotherapies that may prevent sudden cardiac death or irreversible cardiac damage, as in the case of valvular heart disease^{2–12}. The field has therefore focused on identifying feasible strategies to screen for SHD at scale. The application of AI to the 12-lead ECG has resulted in a tool capable of detecting a wide range of SHDs with a high sensitivity, allowing for ruling out the disease in those with a negative AI ECG^{13–16,18–20}. However, the limitations of AI-ECG as a screening tool include modest specificity and, in low-prevalence settings, low positive predictive value (PPV), which would reduce the need for referring all AI-ECG positive patients for comprehensive echocardiography. Given the high clinical demand and cost of diagnostic echocardiography, there is a need for an intermediate, scalable test that can more efficiently identify which patients with an abnormal AI-ECG should proceed to a formal echocardiogram.

Table 4 | Diagnostic performance of AI-ECG alone, AI-FoCUS for all, and a two-step AI-ECG → AI-FoCUS strategy for the composite structural heart disease (SHD) endpoint (in subset of patients with feasible AI-FoCUS)

Metric	AI-ECG alone	AI-FoCUS for all	Two-step (AI-ECG → AI-FoCUS)	AI-FoCUS vs AI-ECG	Two-step vs AI-FoCUS
Analysis set, <i>n</i>	968	968	968		
Composite SHD prevalence	161/ 968 (16.6%)	161/968 (16.6%)	161/968 (16.6%)		
Sensitivity	0.870	0.901	0.770	+0.031	−0.130
Specificity	0.607	0.991	0.994	+0.384	+0.002
Positive predictive value	0.306	0.954	0.961	+0.648	+0.007
Negative predictive value	0.959	0.980	0.956	+0.021	−0.024
Positive likelihood ratio	2.15	103.83	126.0	+101.68	+22.17
Negative likelihood ratio	0.23	0.10	0.23	−0.13	+0.13
Accuracy	0.651	0.976	0.957	+0.325	−0.020
Positive screen rate (TP + FP)	0.472	0.157	0.133	−0.315	−0.024
AI-FoCUS exams performed, <i>n</i> (%)	0 (0.0%)	968 (100.0%)	457 (47.2%)	+968	−511
Incremental AI-FoCUS exams per SHD detected ^a	—	—	—	193.6	24.3
Categorical NRI (2-category) ^b	—	—	—	0.415 (event +0.031; non-event +0.384)	−0.128 (event −0.130; non-event +0.002)

All metrics are calculated on a common cohort (*n* = 968) to allow head-to-head comparisons (i.e., excluding those 27 patients in whom AI-FoCUS was not feasible). The two-step performance corresponds to applying AI-FoCUS only when ≥ 1 AI-ECG algorithm is positive; therefore, the number of AI-FoCUS exams performed in the two-step strategy equals the number of AI-ECG positive screens in this cohort (*n* = 457).

^a Incremental AI-FoCUS exams per structural heart disease (SHD) detected are reported for the direction indicated by the column header. For “AI-FoCUS vs AI-ECG”, this represents the additional number of AI-FoCUS exams per additional SHD case detected when performing AI-FoCUS for all subjects instead of performing AI-ECG alone. For “Two-step vs AI-FoCUS”, the resulting positive ratio represents AI-FoCUS additional exams performed per additional SHD detected when using the two-step strategy instead of AI-FoCUS for all.

^b Categorical NRI (Net Reclassification Index) is calculated for the binary categories (positive vs negative test) on the same cohort: $NRI = (\Delta Sensitivity) + (\Delta Specificity)$. Event and non-event components are shown in parentheses.

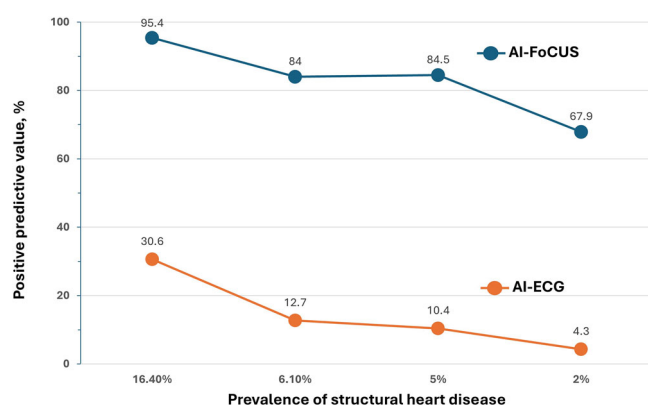


Fig. 5 | Positive predictive values (PPVs) of the AI-ECG and AI-FoCUS screening pathways at various prevalences of structural heart disease. Shown are the measured PPVs in the overall cohort at 16.4% and in those without known structural heart disease (6.1%), and then simulated PPVs at prevalences of 5% and 2%.

Here in this prospective study of 995 adults undergoing clinically indicated echocardiography, we evaluated the ability of AI-ECG and AI-FoCUS performed by novices to detect four forms of SHD. Our results confirmed that the AI-ECG, while having excellent sensitivity for the detection of SHD, the technology in isolation is limited by poor specificity. AI-FoCUS obtained by novices with results reviewed by experts is a superior screening strategy for SHD with very high sensitivity (90.1%), specificity (99.1%), and positive and negative predictive values (95.4% and 98.0%), but requires FoCUS imaging in all. AI-FoCUS also accurately reclassifies most AI-ECG false positives. This underscores the complementary roles of the two modalities: AI-ECG offers broad, sensitive initial screening, while AI-FoCUS effectively eliminates most false positives and hence could be

positioned as a potential interim screen between the AI-ECG and a diagnostic clinical echocardiogram.

The simulated two-step strategy tested in this study of initial AI-ECG followed by AI-FoCUS only in those with at least one positive AI-ECG algorithm provides a pragmatic compromise between diagnostic performance and resource usage. In this framework, only 47% of patients would undergo AI-FoCUS, yet the combined strategy maintains a high PPV (96.1%) and NPV (95.6%) for the composite SHD endpoint, with specificity of 99.4% and sensitivity of 77.0%. Compared with AI-FoCUS for all strategy, this approach reduces the number of handheld ultrasound exams and downstream comprehensive echocardiograms while preserving high accuracy among those ultimately referred. However, this approach occurs at the expense of lower sensitivity, with a modest number of significant cases of SHD missed, including 6 cases of severe AS.

Ultimately, the choice of screening modality for SHD would likely vary between institutions and screening locations, which have differences in resource availability. In resource-rich settings with ample handheld devices and personnel trained in AI-guided FoCUS, performing AI-FoCUS in all patients would likely be preferable due to the superior diagnostic accuracy over both the AI-ECG screening and combined 2-step diagnostic processes. In contrast, in settings where handheld ultrasound devices or trained operators are limited, a two-step pipeline using AI-ECG to initially gate access to AI-FoCUS may offer an attractive balance between sensitivity and resource utilization.

Ultrasound has the capacity to answer simple diagnostic questions in real-time in a host of clinical care settings, and advances in technology have provided the capacity for images to be obtained on handheld devices. However, a major barrier to handheld imaging is operator variability. There is emerging evidence that AI has significant potential in guiding less experienced users to acquire diagnostic-quality ultrasound images. A multicenter validation study demonstrated that healthcare professionals with minimal lung ultrasound experience, when assisted by AI, were able to obtain “expert-level” diagnostic images²¹. The AI system provided real-time guidance during image acquisition, automatically captured clips once

Fig. 6 | Models of the three testing strategies as screening tests and referrals that would be generated for echocardiography. The first using the AI-ECG with a high negative predictive value (NPV) but low positive predictive value (PPV) and was abnormal in 47.0% of cases. The second, where all patients undergo AI-FoCUS, has high PPV and NPV and has a lower abnormality rate/rate of referral for echo, and finally the two-step where only those patients with abnormal AI-ECG (48%) undergo an AI-FoCUS. This would generate the lowest referral for the echocardiogram and is less resource intensive than AI-FoCUS in all patients. Created in BioRender. Kane, G. (2026).

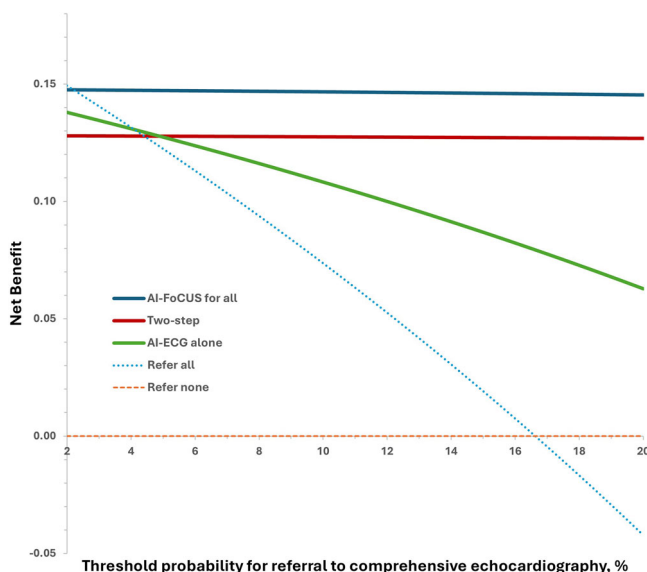
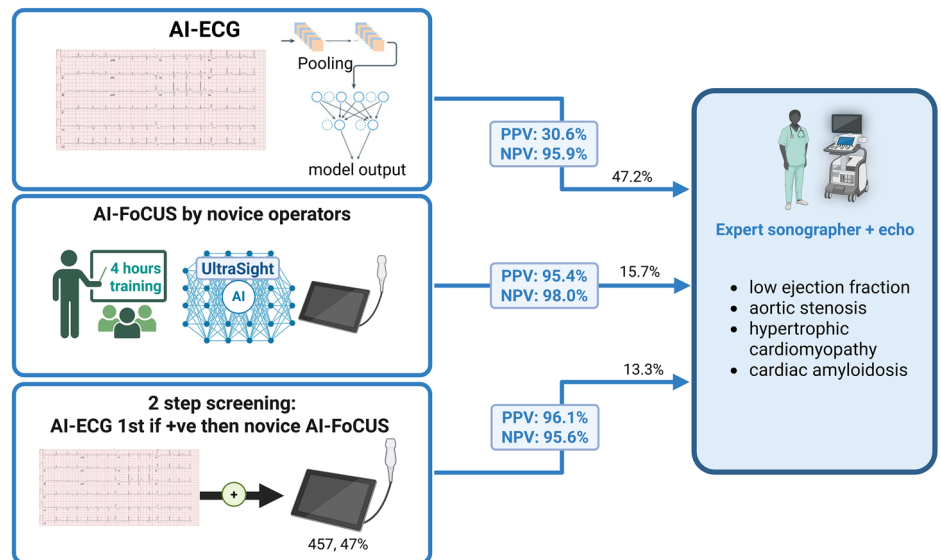


Fig. 7 | Decision curve analysis for the composite structural heart disease endpoint. Standard decision curve analysis showing net benefit for referral to comprehensive echocardiography across threshold probabilities of 2% to 20% for AI-ECG alone, AI-FoCUS for all, and a two-step AI-ECG then AI-FoCUS strategy, together with the default strategies of referring all patients or referring none. Analyses were performed in the common interpretable cohort ($n = 968$), excluding 27 participants with uninterpretable AI-FoCUS studies. Among the three tested screening strategies, AI-FoCUS for all showed the highest net benefit across the plotted threshold range.

quality thresholds were met, and annotated key features such as B-lines²¹. Almost all of the AI-assisted studies met diagnostic quality standards, comparable to those obtained by expert sonographers²¹. A separate study evaluating gestational age estimation found that novice users with no prior sonography training, using an AI-enabled device, achieved accuracy equivalent to credentialed sonographers using traditional equipment²². Other studies have evaluated cardiac imaging, specifically reporting the ability to assess LV ejection fraction by novice users with good diagnostic quality^{17,23–25}. However, unlike in our study, the *novice* users in these studies may be better described as *less experienced*, as they are often nurses or cardiology fellows who had experience in pathology, cardiac views, or prior ultrasound scanning^{23,24}. Some studies have been performed on cart-based

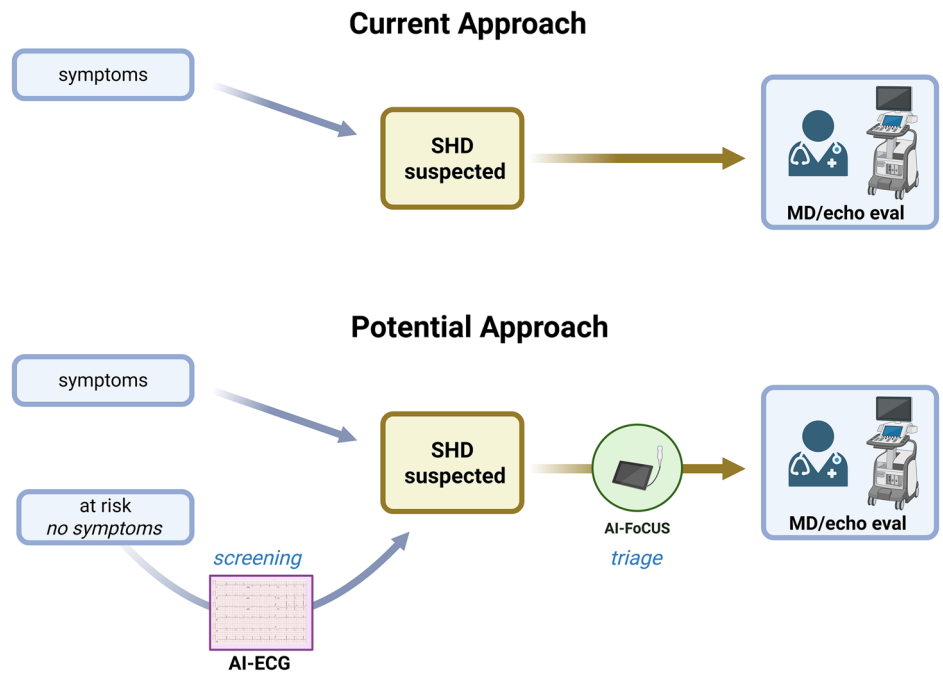
rather than handheld equipment^{17,23,24} and frequently have exclusion criteria for subject recruitment based on image quality, patient position, or body size^{17,25}. Here, we evaluate truly novice users by studying individuals with no clinical experience or imaging training apart from the four-hour introductory workshop explaining the device, use of the software for imaging, patient position, location of the appropriate acoustic window, and the setting of the appropriate gain, depth, and use of ultrasound gel. In addition, patient characteristics such as body size or image quality were not factors in patient selection.

An important clarification is that, in the present study, AI was used solely for image acquisition guidance in the AI-FoCUS workflow, with all diagnostic interpretation of FoCUS images being performed by expert echocardiographers blinded to AI-ECG and clinical data. As such, AI-FoCUS as implemented here does not yet represent a fully automated imaging solution, and the performance we report reflects the combination of novice acquisition plus expert interpretation. The higher performance of AI-FoCUS relative to AI-ECG should be interpreted in the context that the AI-FoCUS pathway combined AI-guided acquisition with expert image interpretation, whereas the AI-ECG pathway was fully automated, which may degrade test performance characteristics. The superior performance of AI-FoCUS relative to AI-ECG is therefore not unexpected, given that echocardiography is the reference imaging modality for SHD and that human experts reviewed the ultrasound images. Nonetheless, our results demonstrate that novice-acquired, AI-guided images can support expert-level diagnosis with excellent reproducibility, an essential prerequisite for future integration with automated interpretation algorithms^{26–29}.

Both AI-ECG and AI-FoCUS have tremendous potential, and our findings underscore several practical considerations for clinical implementation. Handheld ultrasound devices and AI-guidance software require initial investment, yet they are significantly less expensive and more portable than conventional echocardiography systems. This portability enables deployment in primary care clinics, community hospitals, and remote locations, which are already well equipped to perform standardized AI-ECG for screening. Training for novice operators in this study was limited to a single standardized 4-h session, indicating that with the AI-guidance software tested here, competency for the acquisition of basic images may be attainable. Integration of both AI-ECG and AI-FoCUS systems into existing clinical workflows and electronic health records is essential. Although formal cost-effectiveness analyses comparing AI-ECG only, AI-FoCUS only, and two-step strategies were beyond the scope of this study, such analyses will be critical for informing policy and reimbursement decisions.

The advent of AI-guided FoCUS together with the potential development of AI tools for diagnostic interpretation of FoCUS images^{26–30} may

Fig. 8 | Depicts current workflow (upper) where when patients develop symptoms suggestive of structural heart disease (SHD), they are referred for a medical evaluation with echocardiography often with a significant time delay due to availability. Integrating the results of the current work and others offer the potential for an alternative hypothetical future pathway (lower) where at risk, asymptomatic, populations get screened with an AI-ECG and those with positive results or patients with SHD symptoms undergo a more rapidly available novice AI-guided focused ultrasound (AI-FoCUS) scan which would triage diagnosis and priority to better determine those who need a medical expert evaluation and diagnostic testing and at what priority. Created in BioRender. Kane, G. (2026).



reduce the burden of having formal interpretation of echocardiographic images by an expert reviewer and opens up the possibility of a new paradigm in patient care (Fig. 8). Although in this study we have used standard 12-lead ECG as a source for the AI, it has previously shown that AI-ECG derived from one-lead smartwatch/smartphone-based devices has reasonable sensitivity and specificity for SHD detection^{31,32}. Taking this a step further, one could even envisage patients performing their own ECG on wearable devices and those with positive SHD findings being sent handheld devices for AI-guided FoCUS examination by family or self-imaged at the home. Data may then remotely be interpreted in a medical center or by AI, and patients in whom the SHD screening is positive could be triaged for need and priority for formal outpatient evaluation. However, this expanded concept of patient/family member scanning was not tested here, and the current study evaluated feasibility in non-clinical novice users, not patients performing their own scans. Future expanded studies would be needed to explore these considerations.

Our study has several limitations. First, it was conducted at a tertiary care academic center among patients undergoing clinically indicated echocardiography for known or suspected cardiovascular disease. While this was needed to have a reference echocardiogram for comparison as the “gold standard” examination, subjects likely had higher pre-test probability of SHD than would be seen in asymptomatic community screening programs, and some had known SHD diagnoses at the time of enrollment. This referral-population design may inflate PPV and overall accuracy estimates relative to those expected in lower-risk populations and introduces spectrum bias. Our cohort also had limited racial and ethnic diversity, and we were underpowered to perform robust performance analyses across racial/ethnic subgroups. Lastly, while we modelled the accuracy of the AI-FoCUS and AI-ECG as screening tests, our actual population all underwent diagnostic echocardiography as part of clinical care. Future work in broader, more diverse, and truly asymptomatic populations is required to establish generalizability and to refine the role of the two-step and AI-FoCUS for all strategies in screening contexts and assess the impact of these strategies on clinical decision making, patient outcomes, or health-care utilization.

Second, although AI-FoCUS was performed by nine novice operators with no formal clinical or sonography training, they were young adults with high technical literacy and volunteered for a research study, which may not fully reflect the range of potential real-world users. While modest differences

in image quality were seen between users, these were not associated with differences in diagnostic accuracy. We also did not design the study to compare scans obtained by different novices in the same subject. It remains uncertain how performance would change over time with experience, or how reproducible acquisitions would be across a larger and more heterogeneous group of novice users. Moreover, although the overall feasibility of AI-FoCUS was high, image quality varied by view, and a small proportion of patients (2.7%) had inadequate diagnostic images to assess for all SHD pathologies. In patients with partial examinations, experts relied on available views to adjudicate SHD status, reflecting real-world FoCUS practice but potentially underestimating disease in some cases.

Third, the study focused on four SHD phenotypes for which validated AI-ECG algorithms are available at our institution; left ventricular systolic dysfunction, moderate or greater aortic stenosis, hypertrophic cardiomyopathy, and cardiac amyloidosis. We did not evaluate other common structural abnormalities such as right ventricular dysfunction, pulmonary hypertension, or mitral and tricuspid regurgitation. Standard comprehensive echocardiography incorporates color and spectral Doppler imaging in addition to 2D B-mode to evaluate these conditions, whereas our FoCUS protocol relied solely on 2D views. Whether simplified imaging with AI-guided FoCUS can accurately detect a broader range of SHD requires further study, and parallel development of AI-ECG and AI-echo algorithms for these indications is ongoing^{26–30}.

Finally, the AI-FoCUS images in this study required interpretation by an expert echocardiographer. Although review time was brief due to the small number of images (<3 min per study, including image-quality grading), reliance on human experts may still limit scalability in regions with severe shortages of interpreting echocardiographers. AI-interpretative algorithms for cardiac ultrasound images, which are in development, have the potential to reduce the reliance on human experts^{26–30}. We also acknowledge that we used standard 12-lead ECGs for this study, and it remains uncertain whether similar results could be achieved using simplified systems, such as a 6-lead or single-lead ECG, which may be more feasible in resource-limited settings. Future experiments may explore whether the different types of ECG could be analyzed by AI to provide the same level of diagnostic information as the 12-lead ECG.

In conclusion, in this echo referral cohort, AI-guided FoCUS acquired by novice users and interpreted by experts was feasible and accurate.

Performing AI-FoCUS in all participants provided the highest sensitivity, whereas a two-step AI-ECG then AI-FoCUS workflow conserved imaging resources at the cost of additional missed SHD cases. The preferred strategy will therefore depend on whether a given setting prioritizes maximal case detection (*AI-FoCUS alone without any AI-ECG*) or conservation of limited imaging resources (*two step with AI-ECG first, then AI-FoCUS if AI-ECG abnormal*).

Methods

Study population

We prospectively enrolled 995 adult patients to undergo AI-FoCUS within 30 days of an AI-ECG and undergo a clinically indicated comprehensive diagnostic echocardiogram for the evaluation of known or suspected cardiovascular disease (echocardiogram performed within 30 min of their AI-FoCUS) (Fig. 1). The only exclusion criteria were complex congenital heart disease or the inability to consent. Specifically, patients were not excluded based on any weight or body mass index threshold or on the presence of poor acoustic windows. The study was conducted at a tertiary care academic medical center among patients undergoing clinically indicated echocardiography ordered for a range of indications (e.g., dyspnea, suspected cardiomyopathy, murmurs, arrhythmias), as listed in Table 1. Some participants did have pre-existing diagnoses of SHD or cardiovascular disease at the time of inclusion (e.g., prior heart failure or known HCM/amyloidosis), with the clinically indicated echocardiogram often obtained for follow-up or further evaluation. Patients with a prior aortic valve replacement were excluded from the AS evaluation but included for the analysis of LVSD, hypertrophic cardiomyopathy, or cardiac amyloidosis. The study was approved by the Mayo Clinic Institutional Review Board (IRB No. 24-005529). All participants provided informed consent for participation in the study.

SHD

For this study, “structural heart disease” refers to four clinically important conditions: (i) left ventricular systolic dysfunction as defined as the presence of an LVEF < 40%³³, (ii) moderate or greater aortic valvular stenosis⁸, (iii) hypertrophic cardiomyopathy¹¹, and (iv) cardiac amyloidosis¹².

AI-ECG

12-lead electrocardiograms were acquired as part of clinical care. AI-ECG data was generated through validated algorithms using solely input data of digital ECG signals. Each AI-ECG algorithm generates a continuous probability score (0–1). A fixed, pre-specified threshold, established in prior derivation and validation studies for each algorithm, was applied to categorize each case as positive or negative for (i) LVEF < 40%, (ii) ≥ moderate aortic stenosis, (iii) hypertrophic cardiomyopathy, and (iv) cardiac amyloidosis as previously described^{13–16}. For the purposes of the analysis of hypertrophic cardiomyopathy and cardiac amyloidosis, if either AI-ECG model was positive, the case was deemed positive, if both models negative, the case was deemed negative.

AI-FoCUS and data analysis

For the purposes of this study AI-FoCUS referred to a FoCUS study that was performed by one of nine novice operators (college students without formal sonography or clinical medical experience), containing six standard cardiac views, each guided in real-time by AI software (UltraSight Ltd. Inc.) which was integrated into a commercial Lumify hand-held ultrasound device (Fig. 2). This FDA-approved AI-software instructs users how to maneuver the probe for each cardiac view while providing an indicator of quality. Prior to the study, novice operators received very focused training in the use of the software for imaging (standardized 4-hour session that covered patient position, location of the appropriate acoustic window, and the setting of the appropriate gain, depth, and use of ultrasound gel). The novice AI-FoCUS operators were not provided with AI-ECG outputs and had minimal clinical information prior to scanning subjects, with their task limited to following the AI guidance to obtain the required views.

No images were interpreted by AI. All AI-FoCUS images were recorded on a secure server for evaluation by an expert (board-certified cardiologists with advanced echocardiography training (e.g., level III) and more than 10 years of experience), blinded to the AI-ECG results, patient history, and to the original clinical echocardiogram images and report. The experts reviewed only the stored FoCUS images. Findings were assessed qualitatively. LVSD was rated as follows: (i) normal LVEF ≥ 50%; (ii) LVEF 35–49% and (iii) LVEF < 35%. The aortic valve was graded as: (i) normal; (ii) less than moderate stenosis or (iii) moderate or greater stenosis. The visual appearance suggestive of hypertrophic cardiomyopathy or cardiac amyloidosis was graded as absent, possible, or present. For the purposes of analysis, the grading of “possible” or “present” for hypertrophic cardiomyopathy or cardiac amyloidosis was combined as “present”. For patients with partial exams (i.e., some views rated “missing”), the expert interpreted SHD status using the available views, consistent with real-world FoCUS practice, and the test was considered interpretable if sufficient views were present to judge the four SHD categories. The ECG and AI-FoCUS results were then compared to the reference standard of the formal diagnostic echocardiogram, performed according to American Society of Echocardiography guidelines³³.

Quality score

All AI-FoCUS images obtained by novice users were also scored for quality by the expert. The best image(s) for each of 6 standard views were scored as: Excellent (*diagnostic quality equivalent for those obtained by an expert sonographer with a diagnostic echocardiography machine*); Good (*less than typical diagnostic echo quality obtained by an expert sonographer; however, all structures in view could be assessed including adequate quality to qualitatively assess relevant structures and function*); Adequate (*while sub-optimal for all cardiac structures in the view, adequate quality to qualitatively assess relevant structures and function*); Poor (*some structures present but sub-optimal for assessment of relevant structures and function*). If no interpretable image from a particular view was present, it was defined as missing.

Statistics

Statistical analyses were performed using JMP 18.0. Descriptive statistics are presented as mean ± standard deviation, median (interquartile range), or count (percentage), as appropriate. Diagnostic performance was summarized using sensitivity, specificity, PPV, NPV, likelihood ratios, and accuracy, each with 95% confidence intervals. For paired comparisons among AI-ECG alone, AI-FoCUS for all, and the two-step AI-ECG, first, then AI-FoCUS strategy, analyses were performed on the common cohort with interpretable AI-FoCUS examinations. Differences in sensitivity and specificity between paired strategies were assessed using McNemar tests in the diseased and non-diseased subsets, respectively. For the continuous AI-ECG outputs, discrimination was assessed using ROC curves and AUC with 95% confidence intervals by DeLong’s method, and calibration was assessed using calibration plots, calibration intercept, calibration slope, and Brier score. Prevalence-adjusted PPV and NPV scenario analyses were calculated using Bayes’ theorem. Decision-curve analysis was used to estimate net benefit for referral to comprehensive echocardiography. Operator-level learning effects were evaluated using mixed-effects models with operator as a random intercept. Two-sided $p < 0.05$ was considered statistically significant. Some figures were created in <https://BioRender.com>.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to institutional research rules but are available from the corresponding author on reasonable request after institutional IRB approval.

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References

1. Luengo-Fernández, R. et al. Economic burden of cardiovascular diseases in the European Union: a population-based cost study. *Eur. Heart J.* **44**, 4752–4767 (2023).
2. Vantrimpont, P. et al. Additive beneficial effects of beta-blockers to angiotensin-converting enzyme inhibitors in the survival and ventricular enlargement (SAVE) study. SAVE Investigators. *J. Am. Coll. Cardiol.* **29**, 229–236 (1997).
3. Exner, D. V., Dries, D. L., Waclawiw, M. A., Shelton, B. & Domanski, M. J. Beta-adrenergic blocking agent use and mortality in patients with asymptomatic and symptomatic left ventricular systolic dysfunction: a post hoc analysis of the Studies of Left Ventricular Dysfunction. *J. Am. Coll. Cardiol.* **33**, 916–923 (1999).
4. Wang, T. J., Levy, D., Benjamin, E. J. & Vasan, R. S. The epidemiology of “asymptomatic” left ventricular systolic dysfunction: implications for screening. *Ann. Intern Med.* **138**, 907–916 (2003).
5. Jong, P., Yusuf, S., Rousseau, M. F., Ahn, S. A. & Bangdiwala, S. I. Effect of enalapril on 12-year survival and life expectancy in patients with left ventricular systolic dysfunction: a follow-up study. *Lancet Lond. Engl.* **361**, 1843–1848 (2003).
6. Taniguchi, T. et al. Initial surgical versus conservative strategies in patients with asymptomatic severe aortic stenosis. *J. Am. Coll. Cardiol.* **66**, 2827–2838 (2015).
7. Gahl, B. et al. Natural history of asymptomatic severe aortic stenosis and the association of early intervention with outcomes: a systematic review and meta-analysis. *JAMA Cardiol.* **5**, 1102–1112 (2015).
8. Vahanian, A. et al. 2021 ESC/EACTS guidelines for the management of valvular heart disease: developed by the Task Force for the management of valvular heart disease of the European Society of Cardiology (ESC) and the European Association for Cardio-Thoracic Surgery (EACTS). *Rev. Esp. Cardiol. Engl. Ed.* **75**, 524 (2022).
9. Génèreux, P. et al. Natural history, diagnostic approaches, and therapeutic strategies for patients with asymptomatic severe aortic stenosis. *J. Am. Coll. Cardiol.* **67**, 2263–2288 (2016).
10. Castaño, A. et al. Unveiling transthyretin cardiac amyloidosis and its predictors among elderly patients with severe aortic stenosis undergoing transcatheter aortic valve replacement. *Eur. Heart J.* **38**, 2879–2887 (2017).
11. Maron, B. J. et al. Diagnosis and evaluation of hypertrophic cardiomyopathy. *J. Am. Coll. Cardiol.* **79**, 372–389 (2022).
12. Maurer, M. S. et al. Patisiran treatment in patients with transthyretin cardiac amyloidosis. *N. Engl. J. Med.* **389**, 1553–1565 (2023).
13. Yao, X. et al. Artificial intelligence-enabled electrocardiograms for identification of patients with low ejection fraction: a pragmatic, randomized clinical trial. *Nat. Med.* **27**, 815–819 (2021).
14. Cohen-Shelly, M. et al. Electrocardiogram screening for aortic valve stenosis using artificial intelligence. *Eur. Heart J.* **42**, 2885–2896 (2021).
15. Grogan, M. et al. Artificial Intelligence-enhanced electrocardiogram for the early detection of cardiac amyloidosis. *Mayo Clin. Proc.* **96**, 2768–2778 (2021).
16. Ko, W. Y. et al. Detection of hypertrophic cardiomyopathy using a convolutional neural network-enabled electrocardiogram. *J. Am. Coll. Cardiol.* **75**, 722–733 (2020).
17. Mor-Avi, V. et al. Real-time artificial intelligence-based guidance of echocardiographic imaging by novices: image quality and suitability for diagnostic interpretation and quantitative analysis. *Circ. Cardiovasc. Imaging* **16**, e015569 (2023).
18. Lim, J. et al. Artificial intelligence-enhanced six-lead portable electrocardiogram device for detecting left ventricular systolic dysfunction: a prospective single-centre cohort study. *Eur. Heart J. Digit. Health* **6**, 476–485 (2025).
19. Antoniades, C. & Chan, K. Scalable screening for structural heart disease: promises from artificial intelligence-electrocardiogram tools. *Eur. Heart J. Digit. Health* **6**, 521–523 (2025).
20. Poterucha, T. J. et al. Detecting structural heart disease from electrocardiograms using AI. *Nature* **544**, 221–230 (2025).
21. Baloesescu, C. et al. Artificial Intelligence-guided lung ultrasound by nonexperts. *JAMA Cardiol.* **10**, 245–253 (2025).
22. Stringer, J. S. A. et al. Diagnostic accuracy of an integrated AI tool to estimate gestational Age from blind ultrasound sweeps. *JAMA* **332**, 649–657 (2024).
23. Narang, A. et al. Utility of a deep-learning algorithm to guide novices to acquire echocardiograms for limited diagnostic use. *JAMA Cardiol.* **6**, 624–632 (2021).
24. Motazedian, P. et al. Diagnostic accuracy of point-of-care ultrasound with artificial intelligence-assisted assessment of left ventricular ejection fraction. *npj Digit. Med.* **6**, 201 (2023).
25. Asch, F. M. et al. Deep Learning-based automated echocardiographic quantification of left ventricular ejection fraction: a point-of-care solution. *Circ. Cardiovasc. Imaging* **14**, e102293 (2021).
26. Anisuzzaman, D. M. et al. A view-flexible deep learning framework for automated analysis of 2D echocardiography. *npj Cardiovasc. Health* **3**, 17 (2026).
27. Anisuzzaman, D. M. et al. Leveraging comprehensive echo data to power Artificial Intelligence models for handheld cardiac ultrasound. *Mayo Clin. Proceed. Dig. Health* **3**, 100194 (2025).
28. Naser, J. A. et al. Artificial intelligence-based classification of echocardiographic views. *Eur. Heart J. Digit. Health* **5**, 260–269 (2024).
29. Kane C. J. et al. Smarter FoCUS: AI-guided focused cardiac ultrasound enables novice detection of left ventricular dysfunction. *Eur Heart J Dig Health* **7**, 5 (2026).
30. Chao, C. J. et al. Echocardiography-Based deep learning model to differentiate constrictive pericarditis and restrictive cardiomyopathy. *JACC Cardiovasc. Imaging* **17**, 349–360 (2024).
31. Attia, Z. I. et al. Prospective evaluation of smartwatch-enabled detection of left ventricular dysfunction. *Nat. Med.* **28**, 2497–2503 (2022).
32. Attia, Z. I. et al. Automated detection of low ejection fraction from a one-lead electrocardiogram: application of an AI algorithm to an electrocardiogram-enabled Digital Stethoscope. *Eur. Heart J. Digit. Health* **3**, 373–379 (2022).
33. Lang, R. M. et al. Recommendations for cardiac chamber quantification by echocardiography in adults: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *J. Am. Soc. Echocardiogr.* **28**, 1–39 (2015).

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Author contributions

M.M.K.: Formal analysis, Investigation, Project administration, Writing—original draft, Writing—review and editing, and Data curation. J.B., C.J.K., J.K., C.G., M.A.G., D.A.S., M.M.C. and J.Z.: Investigation, Project administration, Writing—review and editing, and Data curation. J.G.M.: Formal analysis, Investigation, Writing—review and editing. G.T., Z.I.A., and J.J.T.: Methodology, Writing—review and editing. E.L., J.A.N., T.J.P., P.A.F., F.L.-J.: Writing—review and editing. G.C.K. and S.V.P.: Conceptualization, Formal analysis, Investigation, Writing—review and editing. J.G.B.: Conceptualization, Formal analysis, Investigation, Project administration, Writing—review and editing, Methodology.

Competing interests

E.L., J.G.M., J.A.N., S.V.P., G.C.K. and Z.I.A. have invented algorithms licensed to UltraSight Ltd. Inc. and may benefit from algorithm commercialization via Mayo Clinic. P.A.F., F.L.-J. and Z.I.A. are members of

the Scientific Advisory Board for Anumana, an AI company commercializing AI-ECG. F.L.-J. is a co-inventor of several algorithms using AI-ECG licensed to Anumana and a co-inventor of an artificial heart valve that has been licensed to Colibri Co. T.J.P. has research grant funds paid to affiliated institutions by Pfizer, Eidos Therapeutics, Janssen, the American Heart Association, and the Tianqiao and Chrissy Chen Foundation, stock ownership in Baxter and Abbott Laboratories, and is an inventor on a patent to use ECG analysis to detect structural heart disease. The remaining authors declare no competing interests.

Additional information

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